

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R3 sends traffic to 192.168.2.0/24 through R1 because R2's S0/0/1 is still passive. The cost 129 comes from: R3 → R1 (64) + R1 → R2 (64) + R2 LAN (1) = 129

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

R3 S0/0/1 to R2 (64) + R2 LAN (1) = 65.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To better represent high-speed links like Gigabit Ethernet. The default 100 Mb/s makes all links ≥ 100 Mb/s have the same cost 1. Increasing it gives more accurate cost differences.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: R1 S0/0/1 (781) + R3 G0/0 (1) = 782

192.168.23.0/30: R1 S0/0/1 (781) + R3 S0/0/1 (64) = 845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. Both serial links now have cost 781. The route goes through two serial links: $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF picks the lowest cost path:

Via R2: $781 + 781 + 1 = 1563$

Via R3: $1565 + 1 = 1566$

$1563 < 1566$, so R1 chooses R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Controlling the router ID prevents network instability. If the router ID is tied to a physical interface and that interface fails, the router ID changes and OSPF must rebuild neighbor relationships. Manual assignment or using a loopback interface keeps the router ID stable and makes the network easier to manage.

2. Why is the DR/BDR election process not a concern in this lab?

Because all links in this lab are point-to-point serial connections. DR/BDR elections only happen on multi-access networks like Ethernet where multiple devices share the same network segment.

3. Why would you want to set an OSPF interface to passive?

To stop sending Hello messages on interfaces that don't need OSPF neighbors, like LAN ports. This saves bandwidth, improves security, and still allows the network to be advertised to other routers.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The ability to divide the network into multiple areas. This limits the spread of routing updates and reduces the processing load on routers, allowing OSPF to work well in large networks.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

`network 10.0.0.0 0.255.255.255 area 0`

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The router compares administrative distance first, not metrics. OSPF has a default AD of 110, RIP is 120, and EIGRP internal is 90. Based on AD, EIGRP would normally be chosen, but the high metric suggests it might be an external route with AD 170. In that case, OSPF 110 would be preferred over RIP 120.